

# Recitation 4 (Section 010)

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September 27, 2026

University of Minnesota

Please scan the QR code and fill out the attendance form!



## Today's Agenda

- Important Announcements
- Last week review
- Auction Experiment
- In Class Activity

## Announcements!

- Midterm is coming up on **October 12th from 6:50pm – 7:50pm.**
- Download both the Honorlock App and Chrome Extension!
- Honorlock Practice Quiz, finish well ahead of midterm!

# Last Week Review

- Price Elasticity of Demand

- A \_\_\_\_\_ increase in the \_\_\_\_\_ leads to a \_\_\_\_\_ in the quantity demanded.
- Is this elasticity positive or negative?

- Income Elasticity of Demand

- A \_\_\_\_\_ increase in the \_\_\_\_\_ leads to a \_\_\_\_\_ in the quantity demanded.
- What does it mean when income elasticity is positive? when negative?

- Cross-Price Elasticity of Demand of Good A relative to Good B.

- A \_\_\_\_\_ increase in the \_\_\_\_\_ of Good \_\_\_\_\_ leads to a \_\_\_\_\_ in the quantity demanded of Good \_\_\_\_\_.
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## Price Elasticity of Demand

How do we calculate Price Elasticity of Demand in this class given datapoints  $(Q_1, P_1)$  and  $(Q_2, P_2)$ ?

A.  $\frac{Q_2 - Q_1}{P_2 - P_1}$

B.  $\frac{(Q_2 - Q_1)/Q_1}{(P_2 - P_1)/P_1}$

C.  $\frac{(Q_2 - Q_1) / \left(\frac{Q_1 + Q_2}{2}\right)}{(P_2 - P_1) / \left(\frac{P_1 + P_2}{2}\right)}$

D.  $\frac{dQ}{dP} \cdot \frac{P_1}{Q_1}$

# Income Elasticity of Demand

How do we calculate Income Elasticity of Demand in this class given datapoints  $(Q_1, I_1)$  and  $(Q_2, I_2)$ ?

A. 
$$\frac{(Q_2 - Q_1) / \left(\frac{Q_1 + Q_2}{2}\right)}{(I_2 - I_1) / \left(\frac{I_1 + I_2}{2}\right)}$$

B. 
$$\frac{(I_2 - I_1) / \left(\frac{I_1 + I_2}{2}\right)}{(Q_2 - Q_1) / \left(\frac{Q_1 + Q_2}{2}\right)}$$

C. 
$$\frac{(Q_2 - Q_1) / Q_1}{(I_2 - I_1) / I_1}$$

D. 
$$\frac{(Q_2 - Q_1) / \left(\frac{Q_1 + Q_2}{2}\right)}{(P_2 - P_1) / \left(\frac{P_1 + P_2}{2}\right)}$$

## Cross-Price Elasticity of Demand

How do we calculate Cross-Price Elasticity of Demand for good  $X$  with respect to the price of good  $Y$ , given datapoints  $(Q_{X,1}, P_{Y,1})$  and  $(Q_{X,2}, P_{Y,2})$ ?

A. 
$$\frac{(Q_{X,2} - Q_{X,1}) / \left( \frac{Q_{X,1} + Q_{X,2}}{2} \right)}{(P_{Y,2} - P_{Y,1}) / \left( \frac{P_{Y,1} + P_{Y,2}}{2} \right)}$$

B. 
$$\frac{(P_{Y,2} - P_{Y,1}) / \left( \frac{P_{Y,1} + P_{Y,2}}{2} \right)}{(Q_{X,2} - Q_{X,1}) / \left( \frac{Q_{X,1} + Q_{X,2}}{2} \right)}$$

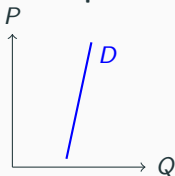
C. 
$$\frac{(Q_{Y,2} - Q_{Y,1}) / \left( \frac{Q_{Y,1} + Q_{Y,2}}{2} \right)}{(P_{X,2} - P_{X,1}) / \left( \frac{P_{X,1} + P_{X,2}}{2} \right)}$$

D. 
$$\frac{(Q_{X,2} - Q_{X,1}) / \left( \frac{Q_{X,1} + Q_{X,2}}{2} \right)}{(P_{X,2} - P_{X,1}) / \left( \frac{P_{X,1} + P_{X,2}}{2} \right)}$$

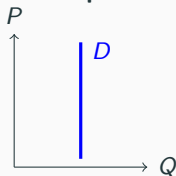
# Identify the Elasticity

For each graph below, identify whether demand is elastic, inelastic, perfectly elastic, perfectly inelastic, or unit elastic over the relevant range.

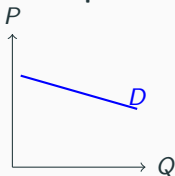
**Graph 1**



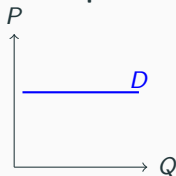
**Graph 3**



**Graph 2**



**Graph 4**



Open up Mindtap!

Open up Canvas and navigate  
to **“In-Class Activity 3  
(Discussions)”**!

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